

Lists, Forms, and Routes, and useParams

1. React Lists

React uses **lists** to render multiple elements dynamically from arrays using the `.map()` method.

Example

```
function StudentList() {  
  const students = ['Ahmed', 'Hasan', 'Turki'];  
  
  return (  
    <ul>  
      {students.map((name, index) => (  

```

2. React Forms

Forms in React use **controlled components** — form inputs are linked to React state.

Example

```
import { useState } from "react";

function ContactForm() {
  const [name, setName] = useState("");

  const handleSubmit = (e) => {
    e.preventDefault();
    alert(`Hello, ${name}!`);
  };

  return (
    <form onSubmit={handleSubmit}>
      <input
        type="text"
        value={name}
        onChange={(e) => setName(e.target.value)}
        placeholder="Enter your name"
      />
    </form>
  );
}
```

Rules

- Always use `value` and `onChange` to control input data.
- Prevent default form submission using `e.preventDefault()`.
- Keep form data in **state** (e.g., with `useState`).

Wrong vs Correct

✗ Wrong (uncontrolled input):

```
<input type="text" placeholder="Name" />
```

✓ Correct (controlled input):

```
<input
  type="text"
  value={name}
  onChange={(e) => setName(e.target.value)}
/>
```

3. React Routes (React Router)

React Router lets you create **multiple pages (routes)** in a single-page React app.

Installation

```
npm install react-router-dom
```

Example

```
import { BrowserRouter, Routes, Route, Link } from "react-router-dom";

function Home() {
  return <h2>Home Page</h2>;
}

function About() {
  return <h2>About Page</h2>;
}

function App() {
  return (
    <BrowserRouter>
      <nav>
        <Link to="/">Home</Link> |{" "}
        <Link to="/about">About</Link>
      </nav>

      <Routes>
        <Route path="/" element={<Home />} />
        <Route path="/about" element={<About />} />
      </Routes>
    </BrowserRouter>
  );
}
```

Rules

- Wrap everything inside `<BrowserRouter>` .
- Use `<Routes>` (not `<Switch>`) for React Router v6+.
- Use `element={<Component />}` instead of `component={Component}` .
- Use `<Link>` instead of `<a>` to avoid full page reload.

Wrong vs Correct

✗ Wrong (old syntax):

```
<Route path="/about" component={About} />  
<a href="/about">About</a>
```

✓ Correct (v6 syntax):

```
<Route path="/about" element={<About />} />  
<Link to="/about">About</Link>
```

4.How useParams Works

Route Definition: You define a route path with placeholders (dynamic segments) marked by a colon (:).

JavaScript

```
<Route path="/products/:productId" element={<ProductDetails />} />
```

URL Matching: When a user navigates to a matching URL (e.g., /products/222), the value (222) is captured.

Extraction: Inside the component (ProductDetails), you call useParams().

Example

If the current URL is **http://localhost:3000/products/222** and the route path is /products/:productId:

Demo [View Demo](#)